

Ryan Cho

Senior Full Stack Software Engineer

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PROFESSIONAL SUMMARY

Senior full stack engineer with 8+ years designing and delivering secure, scalable applications across financial services and enterprise platforms. Expertise in Java, Spring Boot, Node.js, TypeScript, React, AWS microservices, PostgreSQL, and RESTful API architecture. Proven technical leader in Agile environments—integrating AI/LLM capabilities, mentoring engineers, and driving CI/CD, containerization, and cloud-native solutions for mission-critical systems.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Architected and delivered **full-stack platform capabilities** with Java, Spring Boot, Node.js, TypeScript, and React, supporting portfolio analytics and client reporting across **AWS-hosted microservices** used by institutional stakeholders.
- Designed **event-driven microservices** and RESTful APIs backed by PostgreSQL, SQS, and S3, enabling reliable ingestion and processing of high-volume financial datasets with **clear service boundaries** and contract-tested integrations.
- Integrated **LLM-powered workflow features** into internal applications for document summarization and analyst assistance, establishing prompt evaluation, guardrails, and monitoring for **responsible enterprise AI adoption**.
- Led **CI/CD modernization** with Docker, automated unit and integration testing, and pipeline gates that shortened release cycles and improved deployment confidence across **multiple production services**.
- Optimized **PostgreSQL performance** through indexing, query tuning, and connection pooling for reporting workloads, reducing latency on critical endpoints serving **multi-tenant financial data**.
- Partnered with product, QA, and DevOps in **Agile delivery** to translate business and compliance requirements into technical designs, maintaining architecture documentation and **operational runbooks**.
- Mentored engineers on **Spring Boot service design**, React component architecture, and secure coding standards, strengthening code review practices and accelerating **team onboarding**.
- Implemented **authentication and authorization controls** with OAuth2 and JWT across services, aligning application security with audit expectations in a **regulated financial environment**.

Technology Analyst, Incoming Technology Analyst, Technology

May/2019 - Apr/2022

Summer Analyst,

BNP Paribas

Remote | Seventh Avenue, New York

- Built and maintained **Java and Spring Framework services** supporting trading and risk operations, delivering RESTful APIs integrated with market data feeds and **downstream reporting systems**.
- Developed **Node.js and TypeScript utilities** with React dashboards for operations teams, improving visibility into batch processing status and **exception resolution workflows**.
- Contributed to **AWS cloud migration** efforts by containerizing legacy components with Docker and deploying services to ECS using **infrastructure-as-code** practices.
- Expanded **automated test coverage** and Jenkins CI/CD pipelines for payment and reconciliation modules, reducing regression risk during **month-end release windows**.
- Designed **PostgreSQL schemas** and SQL optimizations for transactional reporting, providing analysts with auditable access to high-volume ledger and **reference data**.
- Collaborated with architects and business analysts in **Agile ceremonies** to refine cross-border banking requirements, producing technical specifications and sequence diagrams for **stakeholder alignment**.
- Diagnosed and resolved **production incidents** across distributed Java services, applying root-cause analysis and durable fixes that improved stability during **peak processing periods**.

- Supported **microservices decomposition** of a monolithic risk module by prototyping service boundaries, API contracts, and Kubernetes deployment patterns for **platform modernization**.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Implemented **React components** and REST integrations for customer-facing workflows, improving form validation, state management, and **mobile-responsive layouts**.
- Built **Node.js API endpoints** with Express and PostgreSQL to support inventory management features used by operations staff across **core business processes**.
- Authored **unit and integration tests** with Jest for payment-related modules, catching regressions early and improving confidence during **QA handoffs**.
- Participated in **Agile sprint delivery**, code reviews, and Git-based workflows while consistently completing assigned stories within **sprint commitments**.
- Helped **Dockerize the development environment**, standardizing local setup for engineers and reducing friction during **team onboarding**.
- Refactored shared **frontend utilities** to reduce duplication across pages, improving maintainability and consistency of **UI behavior**.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Developed **responsive web pages** with HTML, CSS, and JavaScript for nonprofit program sites, improving accessibility and **cross-browser compatibility**.
- Integrated **third-party APIs** for newsletter signup and event registration, enabling staff to manage outreach without **manual data entry**.
- Collaborated with stakeholders to ship **content and performance updates**, reducing page load times through asset optimization and **caching strategies**.
- Maintained **Git-based deployment workflows** that supported timely releases for fundraising campaigns and **program launch deadlines**.
- Resolved **front-end defects** reported by users across devices, improving reliability of donation flows and **event registration experiences**.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: React, TypeScript, JavaScript, HTML5, CSS3, Angular, Responsive UI, Component Architecture

Backend: Java, Java 17, Spring Boot, Spring Framework, Node.js, Express, RESTful APIs, Microservices, Event-Driven Architecture

Databases: PostgreSQL, Aurora PostgreSQL, SQL, Schema Design, Query Optimization, Indexing

Cloud & DevOps: AWS (ECS, SQS, S3, Aurora), Docker, Kubernetes, CI/CD, Jenkins, GitHub Actions, Infrastructure as Code

Monitoring & Tools: CloudWatch, JUnit, Jest, Postman, Swagger/OpenAPI, Git, Automated Testing

Workflow & PM: Agile/Scrum, Jira, Confluence, Technical Documentation, Cross-Functional Collaboration, Mentorship